# #3 【学術論文】 Effects of Single-Session High-Definition Transcranial Alternating Current Stimulation (HD-tACS) Combined with Robot-Assisted Hand Exercises on Corticospinal Excitability in Chronic Stroke


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## 1. 論文・ニュース概要

None

## 2. 公開ステータス

- オープンアクセス区分: 有料購読・抄録公開
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## 3. 英文アブストラクト（原文抄録）


OBJECTIVE: This preliminary study investigated whether a single session of high-definition transcranial alternating current stimulation (HD-tACS) could improve corticospinal tract (CST) excitability compared with sham stimulation in healthy individuals and individuals with chronic stroke.

DESIGN: Twelve healthy adults and 7 individuals with chronic stroke received active and sham HD-tACS over the primary motor cortex (M1) flexor digitorum superficialis (FDS) hotspot, ipsilesional in stroke subjects, for 30 minutes at 2mA and 21 Hz in a randomized crossover design. All subjects performed robot-assisted grasp-and-open hand exercises during active and sham stimulation.

RESULTS: HD-tACS significantly increased FDS MEP amplitudes in the healthy group (15.3%). In the stroke group, FDS MEP amplitudes were increased by ~22.3% after HD-tACS, but this change was not statistically significant. When MEP amplitudes were normalized as percentage change, HD-tACS significantly increased FDS MEP amplitudes in both groups compared with the sham condition. No significant extensor digitorum communis (EDC) modulation was observed in either group.

CONCLUSION: HD-tACS combined with robot-assisted hand exercises may enhance CST excitability compared with robot-assisted hand exercises alone after a single session in both healthy and stroke subjects. However, larger studies are needed to confirm these findings.


## 4. 原文確認のご案内

研究の詳細データや査読全文については、必ず上記原文URLよりご確認ください。


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